

===== Start of q1.c =====

//WAP to input numbers into a 3x4 integer array and print evens' as 0 and odds' as 1

```
#include <stdio.h>
```

```
int main(){
    int a[3][4];

    printf("Enter the numbers for the array::\n");
    for (int i=0; i<3; i++){
        for (int j=0; j<4; j++){
            printf("a[%d][%d]", i+1, j+1);
            scanf("%d", &a[i][j]);
        }
    }

    printf("Original Matrix::\n");
    for (int i=0; i<3; i++){
        for(int j=0; j<4; j++){
            printf("\t%d",a[i][j]);
        }
        printf("\n");
    }
    printf("\n\nReplacing Even Numbers with 0 and Odd numbers with 1::\n\n");
    for (int i=0; i<3; i++){
        for(int j=0; j<4; j++){
            if (a[i][j]%2==0){
                printf("0\t");
            }
            else
                printf("1\t");
        }
        printf("\n");
    }
    return 0;
}
```

=====

===== Start of q2.c =====

//WAP to print your full name with a message

```
//
```

```
#include<stdio.h>
```

```
#include<string.h>
```

```
int main(){
    char a[25];
    printf("Enter your name::");
    scanf("%s",a);
    printf("Hello %s\n",a);
}
```

```
        return 0;
    }
```

=====

===== Start of q3.c =====

//Write a program to reverse the elements of an array.

```
#include<stdio.h>
int main(){
    int a[3][4], assign;
    printf ("Enter a number to assing values serially\n");
    scanf("%d",&assign);
    for (int i=0; i<3; i++){
        for(int j=0; j<4; j++){
            a[i][j]= assign;
            assign++;
        }
    }
    printf("\nOriginal Matrix::\n");
    for (int i=0; i<3; i++){
        for(int j=0; j<4; j++){
            printf("%d\t", a[i][j]);
        }
        printf("\n");
    }
    printf("\nReverse Matrix::\n");
    for (int i=2; i>=0; i--){
        for(int j=3; j>=0; j--){
            printf("%d\t", a[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

=====

===== Start of q4.c =====

//Write a program to check whether a matrix is symmetric.

```
#include<stdio.h>
```

```
int main(){
    int a[3][3], b[3][3];

    printf("Enter the numbers for the array::\n");
    for (int i=0; i<3; i++){
```

```

        for (int j=0; j<3; j++){
            printf("a[%d][%d]", i+1, j+1);
            scanf("%d", &a[i][j]);
        }
    }
    for (int i=0; i<3; i++){
        for(int j=0; j<3; j++){
            b[i][j]= a[j][i];
        }
    }
    int count=0;
    for (int i=0; i<3; i++){
        for(int j=0; j<3; j++){
            if (a[i][j]==b[i][j])
                count++;
        }
    }
    for (int i=0; i<3; i++){
        printf("\n");
        for(int j=0; j<3; j++){
            printf("%d\t",a[i][j]);
        }
    }

    if (count==9)
        printf("is symmetric");
    else
        printf(" is not symmetric");

    return 0;
}
=====

```

===== Start of q5.c =====

//Write a program to merge two arrays into a third array

```
#include<stdio.h>
```

```

int main(){
    int a[3][4], b[3][4],c[3][8], assign;
    printf ("Enter a number to assing values serially\n");
    scanf("%d",&assign);
    for (int i=0; i<3; i++){
        for(int j=0; j<4; j++){
            a[i][j]= assign;
            b[i][j]=assign+10;
            assign++;
        }
    }
}

```

```

        for (int i=0; i<3; i++){
            for(int j=0; j<4; j++){
                c[i][j]=a[i][j];
                c[i][j+4]=b[i][j];
            }
        }
        printf("Matrix A:\n");
        for (int i=0; i<3; i++){
            for(int j=0; j<4; j++){
                printf("\t%d",a[i][j]);
            }
            printf("\n");
        }
        printf("Matrix B:\n");
        for (int i=0; i<3; i++){
            for(int j=0; j<4; j++){
                printf("\t%d",b[i][j]);
            }
            printf("\n");
        }
        printf("Merged Matrix:\n");
        for (int i=0; i<3; i++){
            for(int j=0; j<8; j++){
                printf("\t%d",c[i][j]);
            }
            printf("\n");
        }
        return 0;
    }
}

```

=====

===== Start of q6.c =====

```
#include <stdio.h>
```

```

int main() {
    int n, arm = 0, remainder, temp;
    printf("Enter a 3-digit Number: ");
    scanf("%d", &n);

    temp = n;

    for (int i = temp; i > 0; i /= 10) {
        remainder = i % 10;
        arm += (remainder * remainder * remainder);
    }

    if (arm == n)
        printf("%d is an Armstrong number.\n", n);
}

```

```

else
    printf("%d is not an Armstrong number.\n", n);

return 0;
}
=====

===== Start of q7.c =====
//Write a program to determine the type of triangle based on sides by taking input.

#include<stdio.h>

int main(){
    int a,b,c;
    printf("Enter all three sides in order of AB(a), BC(b) and CA(c)::");
    scanf("%d%d%d", &a, &b, &c);
    if (a+b>c && a+c>b && b+c>a){
        if (a==b && b==c)
            printf("ABC is a Equilateral Triangle");
        else if(a==b || a==c || b==c)
            printf("ABC is an Isosceles Triangle");
        else
            printf("ABC is a Scalene Triangle");
    }
    else
        printf("This triangle is not possible.....");
    return 0;
}

=====

===== Start of q8.c =====
//Write a program to generate the Fibonacci series up to n terms.

#include<stdio.h>

int main(){
    int a=0, b=1, c, n;
    printf("Fibonacci series upto::");
    scanf("%d",&n);

    for (int i=1; i<=n; i++){

        printf("%d\t", a);
        c=a+b;
        a=b;
        b=c;
    }
    printf("\n");
}

```

```
        return 0;
    }
```

```
=====
```

```
===== Start of q9.c =====
```

```
//Write a program to print all prime numbers between two given numbers
```

```
//
```

```
#include<stdio.h>
```

```
int main(){
    int start, end, factor;
    printf("\nStart:");
    scanf("%d",&start);
    printf("\nEnd:");
    scanf("%d", &end);

    for (int i=start; i<end; i++){
        factor=0;
        for (int j=1; j<i; j++){
            if (i%j==0){
                factor++;
            }
        }
        if (factor>2)
            printf("-\t");
        else
            printf("%d\t", i);
    }
    return 0;
}
```

```
=====
```

```
===== Start of q10.c =====
```

```
//
```

```
#include <stdio.h>
```

```
int main(){
    int count=0, n;
    printf("Enter a Number::");
    scanf("%d", &n);
    int temp=n;
    if(n==0)
```

```

        count=1;
    else{if(n<0)
        n=-n;
        while (n>0){
            count++;
            n/=10;
        }
        printf("\nThe number of digits in %d is %d\n", temp, count);
        return 0;
    }
}

```

=====

===== Start of q11.c =====

//Write a program to print all Armstrong numbers between 1 and 1000.

//

#include <stdio.h>

#include <math.h>

int main() {

int num, temp, digit;

double sum;

printf("Armstrong numbers between 1 and 1000 are:\n");

for (num = 1; num <= 1000; num++) {

temp = num;

sum = 0;

while (temp > 0) {

digit = temp % 10;

sum += pow(digit, 3);

temp /= 10;

}

if ((int)sum == num) {

printf("%d ", num);

}

}

printf("\n");

return 0;

}

=====

===== Start of q12.c =====

//Write a program to find the GCD and LCM of two numbers.

//

```

#include <stdio.h>

int main() {
    int a, b, x, y, gcd, lcm;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    x = a;
    y = b;

    while (y != 0) {
        int temp = y;
        y = x % y;
        x = temp;
    }

    gcd = x;
    lcm = (a * b) / gcd;

    printf("GCD of %d and %d = %d\n", a, b, gcd);
    printf("LCM of %d and %d = %d\n", a, b, lcm);

    return 0;
}

```

=====

```

===== Start of q13.c =====
#include <stdio.h>

```

```

int main() {
    int dec;
    long long binary = 0;
    long long place = 1;

    printf("Enter an integer [Decimal/Base 10]: ");
    scanf("%d", &dec);

    int temp = dec;

    if (temp == 0) {
        binary = 0;
    } else {
        while (temp > 0) {
            int remainder = temp % 2;
            binary += remainder * place;
            temp /= 2;
            place *= 10;
        }
    }
}

```



```

    }
}

printf("(%d)\u2081\u2080 is (%lld)\u2082\n", dec, binary);

return 0;
}
=====

```

```

===== Start of q14.c =====
//Write a program to print the following pyramid using * as,
//  *
// ***
// *****
// *****
//*****

```

```

#include <stdio.h>

int main() {
    int rows = 5;

    for (int i = 1; i <= rows; i++) {
        for (int space = 1; space <= rows - i; space++) {
            printf(" ");
        }
        for (int j = 1; j <= (2 * i - 1); j++) {
            printf("*");
        }
        printf("\n");
    }

    return 0;
}
=====

```

```

===== Start of q15.c =====
/*
Write a program to print an inverted full pyramid using * as
*****
*****
*****
***
*

*/

```

```

#include <stdio.h>
int main() {

```

```

int rows = 5;

for (int i = rows; i >= 1; i--) {
    for (int space = 1; space <= rows - i; space++) {
        printf(" ");
    }
    for (int j = 1; j <= (2 * i - 1); j++) {
        printf("*");
    }
    printf("\n");
}

return 0;
}
=====

```

===== Start of q16.c =====

```

/*
Write a program to the following pattern,

```

```

1
121
12321
1234321
123454321
*/

```

```

#include <stdio.h>

```

```

int main() {
    int rows = 5;
    for (int i = 1; i <= rows; i++) {
        for (int space = 1; space <= rows - i; space++) {
            printf(" ");
        }
        for (int j = 1; j <= i; j++) {
            printf("%d", j);
        }
        for (int j = i - 1; j >= 1; j--) {
            printf("%d", j);
        }
        printf("\n");
    }
    return 0;
}
=====

```